Narendra Sir
Biology Lecturer | Founder & CEO

TOPIC 8: MENDELIAN AND CHROMOSOMAL GENETIC DISORDERS

PU-II Principles of Inheritance and Variation • Topic-Wise Worksheet

Student Name: ____ **Roll / Reg. No:** ____________ **Date / Batch:** _________

SECTION A: Multiple Choice Questions (1 Mark Each)

1. Down's syndrome is a chromosomal disorder caused by:

(A) Trisomy of chromosome 21 (2n+1 = 47)
(B) Monosomy of X (45, XO)
(C) Trisomy of sex chromosomes (47, XXY)
(D) Deletion of short arm of chromosome 5

2. A human female with karyotype 45, XO having rudimentary ovaries and webbed neck suffers from:

(A) Klinefelter's syndrome
(B) Turner's syndrome
(C) Down's syndrome
(D) Cri-du-chat syndrome

3. Alpha-thalassemia is governed by two closely linked genes located on human chromosome:

(A) Chromosome 11 (HBB)
(B) Chromosome 16 (HBA1 & HBA2)
(C) Chromosome 12
(D) Chromosome 21

4. Haemophilia A ('Royal disease') is inherited as an:

(A) Autosomal dominant trait
(B) Autosomal recessive trait
(C) X-linked recessive trait
(D) Y-linked trait

SECTION B: Fill in the Blanks (1 Mark Each)

5. Klinefelter's syndrome individuals have karyotype 47 with sex chromosome constitution ....

6. The French syndrome 'Cri-du-chat' is caused by partial deletion of the short arm of chromosome ....

7. Hypertrichosis (hairy ears) is transmitted exclusively from father to son as a .... linked trait.

8. Phenylketonuria results from deficiency of the liver enzyme ....

SECTION C: Very Short Answer Questions (2 Marks Each)

9. **[2M]** Differentiate between Aneuploidy and Polyploidy.

10. **[2M]** What is Gynaecomastia? In which chromosomal disorder is it typically observed?

SECTION D: Short Answer Questions (3 Marks Each)

11. **[3M]** Explain why haemophilia is rarely observed in human females compared to males.

12. **[3M]** Explain the biochemical cause and key clinical symptoms of Phenylketonuria (PKU).

SECTION E: Long Answer / Essay Questions (5 Marks Each)

13. **[5M]** Compare the three major human chromosomal disorders (Down's syndrome, Klinefelter's syndrome, Turner's syndrome) regarding: genetic cause, karyotype, chromosome count, and characteristic symptoms.

14. **[5M]** Describe the inheritance, molecular cause, and pathological consequences of: (a) Sickle Cell Anaemia (b) Thalassemia (alpha and beta types).